

Quantum field theory cheat sheet

– prerequisites –

Hitoshi Inamori

June 7, 2026

Abstract

From David Tong Quantum Field Theory Part III Mathematical Tripos course and Peskin Schroeder “An introduction to Quantum Field Theory”

Mathematical preliminaries

Fourier transform, Dirac functions	
(D) Fourier transform (one dimension)	$f(x) = \int_{-\infty}^{+\infty} \frac{dk}{2\pi} e^{-ikx} \tilde{f}(k)$ $\tilde{f}(k) = \int_{-\infty}^{+\infty} dx e^{+ikx} f(x)$
(P) Dirac function	$\int dx f(x) \delta(x - a) = f(a)$ $\int dx f(x) \delta^{(n)}(x - a) = (-1)^n f^{(n)}(a)$ $\delta(g(x)) = \sum_i \frac{\delta(x - x_i)}{ g'(x_i) } \text{ where } x_i \text{ are roots of the function } g$ $\delta(\vec{g}(\vec{x})) = \sum_i \frac{\delta(\vec{x} - \vec{x}_i)}{\det \left(\frac{\partial g_i}{\partial x_j} \right)_{ij}} \text{ where } \vec{x}_i \text{ are roots of the function } \vec{g}$ $\int_{-\infty}^{+\infty} dx e^{ikx} = \int_{-\infty}^{+\infty} dx e^{-ikx} = (2\pi) \delta(k)$
Contour integral	
(P) Darboux inequality	If $f(z) \leq M$ for any point z on a contour C (not necessarily a loop), then $\left \int_C f(z) dz \right \leq ML$ where L is the length of the contour C.
(P) Residue theorem (simplified)	Let $\gamma : [a, b] \rightarrow \mathbf{C}$ be a positively oriented loop i.e. $\gamma(t)$ runs over C anti-clockwise (trigonometric wise). Let f be analytic inside C except for a finite number of poles of order 1 $\{z_i\}_i$ inside C (meaning $(z - z_i)f(z)$ is analytic at z_i). Then $\int_C f(z) dz = 2\pi i \sum_i \text{Res}(f, z_i)$ where $\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} [(z - z_0)f(z)]$

Other mathematical results	
(P) Riemann-Lebesgue lemma	Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a function. If the Lebesgue integral of $ f $ is finite then $\int_{-\infty}^{\infty} dx f(x) e^{-izx} \rightarrow 0 \text{ when } z \rightarrow \infty$
Pauli matrices	
(D) Pauli matrices	$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ $\sigma^i \sigma^j = \delta^{ij} \mathbf{1} + i \epsilon^{ijk} \sigma^k \text{ where } \epsilon^{ijk} \text{ is antisymmetric and } \epsilon^{123} = 1$ $\{\sigma^i, \sigma^j\} = \sigma^i \sigma^j + \sigma^j \sigma^i = 2\delta^{ij} \mathbf{1}$ $[\sigma^i, \sigma^j] = \sigma^i \sigma^j - \sigma^j \sigma^i = 2i \epsilon^{ijk} \sigma^k$
(D) Pauli matrices for spinors	$\sigma^\mu = (\mathbf{1}, \sigma^k) \quad \bar{\sigma}^\mu = (\mathbf{1}, -\sigma^k)$
Clifford algebra	
(D) Clifford algebra	A set of four matrices γ^μ $\mu = 0, 1, 2, 3$ form a Clifford algebra if $\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \mathbf{1}$
(P) Weyl/Chiral representation	Clifford algebra requires at least 4 dimension (4×4 matrices). The following matrices, called Weyl or Chiral representation, form a Clifford algebra: $\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}$ Properties: (1) $(\gamma^0)^2 = \mathbf{1}, (\gamma^k)^2 = -\mathbf{1}$ (2) $(\gamma^0)^\dagger = \gamma^0, (\gamma^k)^\dagger = -\gamma^k$ (3) $\gamma^0 \gamma^\mu \gamma^0 = (\gamma^\mu)^\dagger$
(D) γ^5	$\gamma^5 = -i \gamma^0 \gamma^1 \gamma^2 \gamma^3$ Properties: (1) $\{\gamma^5, \gamma^\mu\} = 0$ (2) $(\gamma^5)^2 = \mathbf{1}$ (3) In the Chiral representation, $\gamma^5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}$
(D) $\not{A}$	For any four vector A^μ , $\not{A} = \gamma_\mu A^\mu$
(P) Trace identities	For Clifford algebra in 4 dimension, (1) The trace of any product of odd number of γ 's is zero (2) $\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu}$ (3) $\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho})$ (4) $\text{Tr}(\gamma^5) = 0$ (5) $\text{Tr}(\gamma^\mu \gamma^\nu \gamma^5) = 0$

Units	
(D) natural units	$\hbar = c = 1$
(P) dimension	The dimension is given by the mass dimension, $[\text{mass}] = 1$. $[\text{length}] = [\text{time}] = -1$ $[\text{energy}] = [\text{mass}] = 1$ $[\text{action}] = 0$ as it is the argument of an exponential function $[\text{Lagrangian density}] = d$ as $S = \int d^d x \mathcal{L}$ From this you get the dimension for the field. Generally renormalizable theory must have dimensionless coupling constants.
(P) few numbers	electron mass 0.5 MeV proton mass 1 GeV

Special relativity

Special relativity	
(D) Minkowski metric	$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$ For any four-vector $X = X^\mu = (X^0, X^1, X^2, X^3) = (X^0, \vec{X})$ the covariant coordinates are $X_\nu = \eta_{\nu\mu} X^\mu = (X^0, -\vec{X})$ (Scalar product) for any four vectors X and Y, $X \cdot Y = X^\mu Y_\mu = \eta_{\mu\nu} X^\mu Y^\nu = X^0 Y^0 - \vec{X} \cdot \vec{Y} = X^0 Y^0 - \sum_k X^k Y^k$ (note that $X \cdot Y$ and $\vec{X} \cdot \vec{Y}$ have opposite sign for the space component) $\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial x^0}, \vec{\nabla} \right) \quad \partial^\mu = \frac{\partial}{\partial x_\mu} = \left(\frac{\partial}{\partial x^0}, -\vec{\nabla} \right)$ $\partial^\mu \partial_\mu = \frac{\partial^2}{\partial (x^0)^2} - \Delta \text{ where } \Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$
(D) Lorentz transformation	Under a Lorentz transformation, the four-vector transform as: $X^\mu \mapsto (X')^\mu = \Lambda^\mu{}_\nu X^\nu$ with the condition $\Lambda^\mu{}_\alpha \eta^{\alpha\beta} \Lambda^\nu{}_\beta = \eta^{\mu\nu}$ The Lorentz transformations form a group (if Λ and Λ' are two Lorentz transformations then $\Lambda\Lambda'$ is a Lorentz transformation). It is composed of: (1) the proper orthochronous Lorentz transformations Λ for which there are 6 real numbers $\Omega_{\rho\sigma}$ with $\Omega_{\rho\sigma} = -\Omega_{\sigma\rho}$ such as $\Lambda = \exp \left[\frac{1}{2} \Omega_{\rho\sigma} \mathcal{M}^{\rho\sigma} \right]$ where $(\mathcal{M}^{\rho\sigma})^{\alpha\beta} = \eta^{\rho\alpha} \eta^{\sigma\beta} - \eta^{\sigma\alpha} \eta^{\rho\beta}$ The $\Omega_{i,j}$ leads to spatial rotations. Example $\Omega_{1,2} = \theta$ leads to $\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ i.e. a rotation in the } xy \text{ plane.}$ The $\Omega_{0,i}$ leads to Lorentz boosts. Example $\Omega_{0,1} = \phi$ leads to $\Lambda = \begin{pmatrix} \cosh \phi & -\sinh \phi & 0 & 0 \\ -\sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ i.e. a speed increase in the } x$ direction, with $\cosh \phi = 1/\sqrt{1-v^2}$ and $\sinh \phi = v/\sqrt{1-v^2}$. (2) the Parity transformation $P = \text{diag}(1, -1, -1, -1)$ (3) the Time reversal transformation $T = \text{diag}(-1, 1, 1, 1)$

(D) Lorentz scalar/vector fields	A Lorentz scalar field or a Lorentz invariant is a scalar field $\phi(x)$ such that under the proper orthochronous Lorentz transformation $x \mapsto \Lambda x$, $\phi(x) \mapsto \phi'(x) = \phi(\Lambda^{-1}x)$ i.e. the transformed field ϕ' at the transformed coordinate Λx is equal to the original field at the original coordinates. A Lorentz vector field is a four-vector field $A(x) = A^\mu(x)$ such that under the proper orthochronous Lorentz transformation $x \mapsto \Lambda x$, $A^\mu(x) \mapsto A'^\mu(x) = \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x)$
(P) Lorentz invariant measures	For any four vector p (1) $d^4p = dp^0 dp^1 dp^2 dp^3 = \left \frac{D(p^\mu)}{D(p'^\mu)} \right dp'^0 dp'^1 dp'^2 dp'^3 = dp'^0 dp'^1 dp'^2 dp'^3$ is a Lorentz invariant (Jacobian is 1). (2) $\delta(p^2 - C)$ and $\theta(p^0 > K)$ are Lorentz invariant, where C and K are constant and $\theta(x)$ is the Heaviside function (3) therefore $\int d^4p \delta(p_0^2 - \vec{p}^2 - C) \theta(p_0 > 0) = \int \frac{d^3\vec{p}}{2p_0} \Big _{p_0=\sqrt{\vec{p}^2+C}}$ is a Lorentz invariant measure, (4) and $2p_0 \delta(\vec{p} - \vec{q})$ is a Lorentz invariant, as $\int \frac{d^3\vec{p}}{2p_0} 2p_0 \delta(\vec{p} - \vec{q}) = 1$ is a Lorentz invariant.

Spinors	
(P) Matrix representation of the Lorentz group	A mapping $\Lambda \mapsto M(\Lambda) \in \mathcal{M}_n(\mathbf{C})$ is a representation of the Lorentz group for the n-dimensional field $\psi(x) \in \mathbf{C}^n$ if, for any Lorentz transformation, $\psi(x) \mapsto \psi'(x) = M(\Lambda)\psi(\Lambda^{-1}x)$ The Lorentz transformations form a group, i.e. $\Lambda\Lambda'$ is Lorentz transformation for any Lorentz transformations Λ and Λ', and we have $M(\Lambda)M(\Lambda') = M(\Lambda\Lambda')$. For any continuous group, the transformations that lie infinitesimally close to the identity define a vector space, called the Lie algebra of the group. The basis vectors for this vector space are called the generators of the Lie algebra, or of the group. For four-vectors, we have 6 generators of the Lie algebra $\mathcal{M}^{\rho\sigma}$ as any proper orthochronous Lorentz transformations Λ reads $\Lambda = \exp \left[\frac{1}{2} \Omega_{\rho\sigma} \mathcal{M}^{\rho\sigma} \right]$ and they obey the Lie algebra relation $[\mathcal{M}^{\alpha\beta}, \mathcal{M}^{\gamma\delta}] = \eta^{\beta\gamma} \mathcal{M}^{\alpha\delta} - \eta^{\alpha\gamma} \mathcal{M}^{\beta\delta} + \eta^{\alpha\delta} \mathcal{M}^{\beta\gamma} - \eta^{\beta\delta} \mathcal{M}^{\alpha\gamma}$ If we have any set of 6 matrices in $\mathcal{M}_n(\mathbf{C})$, $S^{\rho\sigma}$ (with $S^{\rho\sigma} = -S^{\sigma\rho}$) which obey the same Lie algebra relations, then the matrices $S^{\rho\sigma}$ form a representation of the Lie algebra and $S[\Lambda] = \exp \left[\frac{1}{2} \Omega_{\rho\sigma} S^{\rho\sigma} \right]$ is the representation of the Lorentz transformation Λ above.
(P) Spinor representation	The 6 matrices in $\mathcal{M}_4(\mathbf{C})$ $S^{\rho\sigma} = \frac{1}{4} [\gamma^\rho, \gamma^\sigma] = \frac{1}{2} \gamma^\rho \gamma^\sigma - \frac{1}{2} \eta^{\rho\sigma} \mathbf{1}$ form a representation of the Lorentz Lie algebra. For a spatial rotation $\Omega_{ij} = -\epsilon_{ijk} \phi^k$ (rotation of angle ϕ around vector $\vec{\phi}$) we have $S[\Lambda] = \exp \left(\frac{1}{2} \Omega_{\rho\sigma} S^{\rho\sigma} \right) = \begin{pmatrix} e^{+i\vec{\phi} \cdot \vec{\sigma}/2} & 0 \\ 0 & e^{+i\vec{\phi} \cdot \vec{\sigma}/2} \end{pmatrix}$ Note that a 2π rotation around axis z gives $S[\Lambda] = -\mathbf{1}$. For a boost $\Omega_{0i} = \chi_i$, we have $S[\Lambda] = \begin{pmatrix} e^{\vec{\chi} \cdot \vec{\sigma}/2} & 0 \\ 0 & e^{-\vec{\chi} \cdot \vec{\sigma}/2} \end{pmatrix}$ Note that it is not unitary ($S[\Lambda]^\dagger S[\Lambda] \neq \mathbf{1}$). There exists no finite dimensional unitary representations for the Lorentz boosts, which is not a compact group. A map $\psi : x \mapsto \psi(x) \in \mathbf{C}^4$ which transforms as $\psi(x) \mapsto S[\Lambda]\psi(\Lambda^{-1}x)$ under Lorentz transformation is a spinor field.
(P) The adjoint spinor	From the identity $\gamma^0 \gamma^\mu \gamma^0 = (\gamma^\mu)^\dagger$ we get $(S^{\mu\nu})^\dagger = -\gamma^0 S^{\mu\nu} \gamma^0$ i.e. $S[\Lambda]^\dagger = \exp \left(\frac{1}{2} \Omega_{\rho\sigma} (S^{\rho\sigma})^\dagger \right) = \gamma^0 S[\Lambda]^{-1} \gamma^0$ Define the adjoint of a spinor field $\psi(x)$ by $\bar{\psi} = \psi^\dagger \gamma^0$ then $\bar{\psi}\psi$ is a Lorentz scalar, and $\left(\bar{\psi} \gamma^\mu \psi \right)_\mu$ a Lorentz vector.

Scattering amplitude, Decay rate, Cross section	
(D) Scattering amplitude	Let $S = U(t_+, t_-)$ be the unitary evolution operator, and let $i\rangle$ be the initial state of total momentum P_i and $f\rangle$ be the final state of total momentum P_f. The transition amplitude $\mathcal{A}_{fi}$ between the states $i\rangle$ and $f\rangle$ is defined by: $\langle f (S - \mathbf{1}) i \rangle = i \mathcal{A}_{fi} (2\pi)^4 \delta(P_f - P_i)$
(P) Transition probability	The probability for particles in initial state $i\rangle$ to evolve into a final state $f\rangle$ (specified by the momenta $\vec{p}_f$) is given by $P = \frac{ \langle f (S - \mathbf{1}) i \rangle ^2}{\langle f f \rangle \langle i i \rangle}$ with the Lorentz invariant normalisations $\langle i i \rangle = \prod_i 2E_{\vec{p}_i} V$ where V stands for $(2\pi)^3 \delta(\vec{0})$, and $\langle f f \rangle = \prod_f 2E_{\vec{p}_f} d^3 p_f (2\pi)^3$. We get $dP = \prod_i \frac{1}{2E_{\vec{p}_i} V} \mathcal{A}_{fi} ^2 VT d\Pi, \text{ with}$ $d\Pi = (2\pi)^4 \delta(P_f - P_i) \prod_f \frac{d^3 p_f}{(2\pi)^3} \frac{1}{2E_{\vec{p}_f}}$ where VT stands for one of the $(2\pi)^4 \delta(P_f - P_i)$.
(P) Decay rate	If the initial state is a single particle at rest, $E_{\vec{p}_i} = m$ and the decay probability per unit of time $\Gamma = \partial_0 P$ reads $\Gamma = \frac{1}{2m} \int \mathcal{A}_{fi} ^2 d\Pi$
(P) Cross section	If the initial state is a set of converging two particles 1 and 2, the differential cross section, defined as the probability of a scattering occurring during an unit time and for an unit flux F is $d\sigma = \frac{1}{2E_1} \frac{1}{2E_2} \frac{1}{VF} \mathcal{A}_{fi} ^2 d\Pi = \frac{1}{4E_1 E_2} \frac{1}{ \vec{v}_1 - \vec{v}_2 } \mathcal{A}_{fi} ^2 d\Pi$ with $F = \vec{v}_1 - \vec{v}_2 /V$.
(D) Mandelstam variable	For the situation with 2 incoming particles of momenta p_1 and p_2 and 2 outgoing particles of momenta p'_1 and p'_2, we define the Mandelstam variables: $s = (p_1 + p_2)^2 = (p'_1 + p'_2)^2$ $t = (p_1 - p'_1)^2 = (p_2 - p'_2)^2$ $u = (p_1 - p'_2)^2 = (p_2 - p'_1)^2$ With the following property: $s + t + u = \sum_i m_i^2$